



Foundational Agent Pipeline for Cloud Infrastructure Configuration

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Summary

- We introduce a **Multi-Agent System (MAS)** framework grounded in a **Cloud Native World Model**. The system leverages **specialized agents** (Config, YAML, Module, Docs) coordinated by a Master Agent to systematically decompose, validate, and synthesize complex cloud configurations.
- Compared with traditional ground-truth comparison, we implement **Dynamic Evaluation** via a local Kubernetes environment, where automated deployment feedback and schema validation act as the evaluation oracle.
- The **Environment-Reflection** mechanism enables agents to learn from system feedback via a verifier-in-the-loop architecture (VERA), propagating system-level configuration error messages to the agent for iterative diagnosis and resolution.
- Evaluated on real-world **Stack Overflow data**, our enhanced MAS pipeline improves configuration constraint retention by **18.6%** over baselines, reduces knowledge-staleness errors, and achieves a **73.3%** win rate in A/B testing for end-to-end deployment success.

Background: Cloud Configuration with LLM Agents

Cloud Configuration Errors. Modern cloud-native systems rely on complex configuration artifacts such as Kubernetes YAML, Helm charts, and Infrastructure-as-Code scripts. Prior work shows that Kubernetes configuration defects are common and can cause deployment failures, security risks, and service outages [Zhang et al., 2025; Rahman et al., 2025].

LLMs for Cloud Operations. Recent AIOps studies explore LLMs for log analysis, root-cause localization, incident diagnosis, and automated remediation [Zhang et al., 2025; Pasquini et al., 2025]. In Kubernetes settings, systems such as KubeLLM use multi-agent collaboration to inspect cluster states, analyze error logs, and propose configuration fixes [De Jesus et al., 2025].

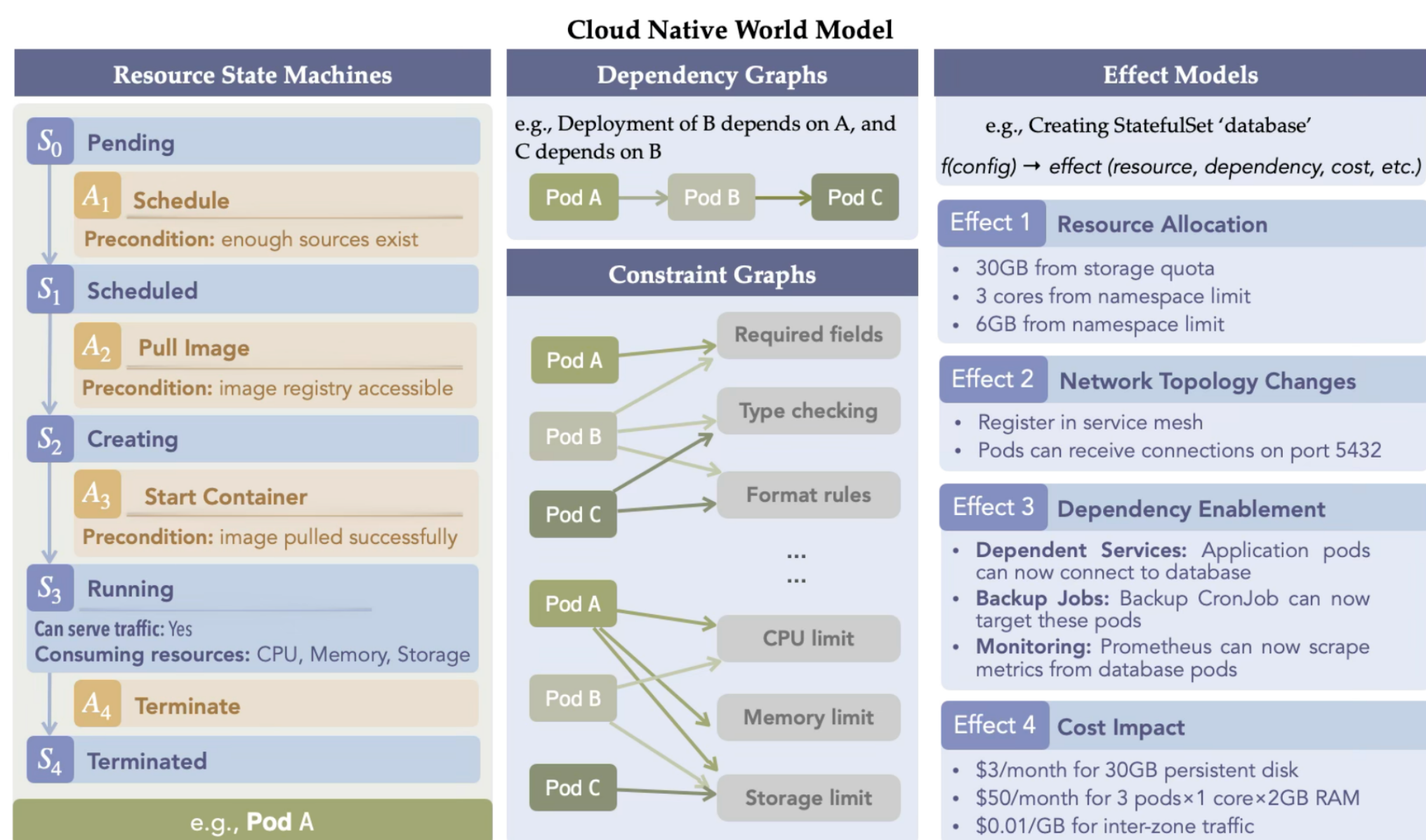
Configuration Repair Paradigms. Existing LLM-based repair methods can be grouped into:

- Generation-based:** translate user intent into Kubernetes or IaC configurations.
- Diagnosis-based:** localize failures from logs, events, and deployment feedback.
- Repair-based:** patch invalid or insecure configurations using validators, static analyzers, or execution feedback.

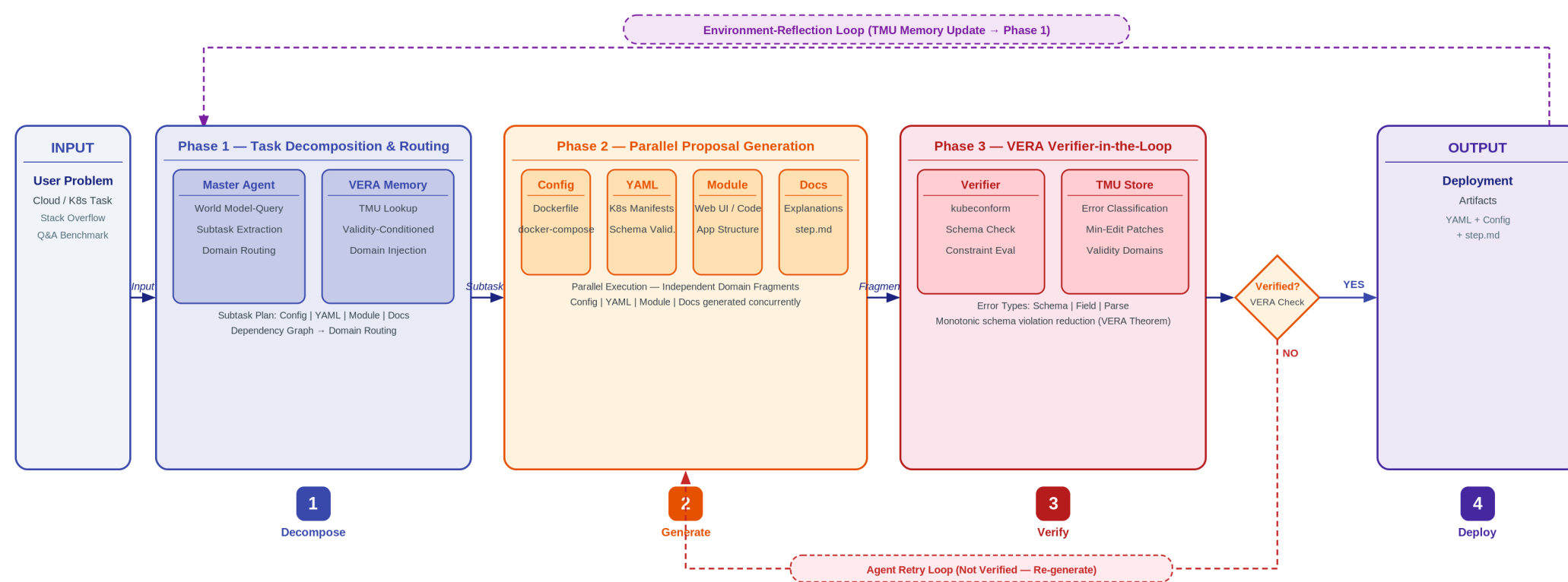
Environment-Reflection MAS for Cloud Configuration

VERA is a multi-agent reasoning framework that exploits **environment-reflection**: it dynamically adapts to deployment feedback and activates specialized agents to resolve **complex cross-layer dependencies**.

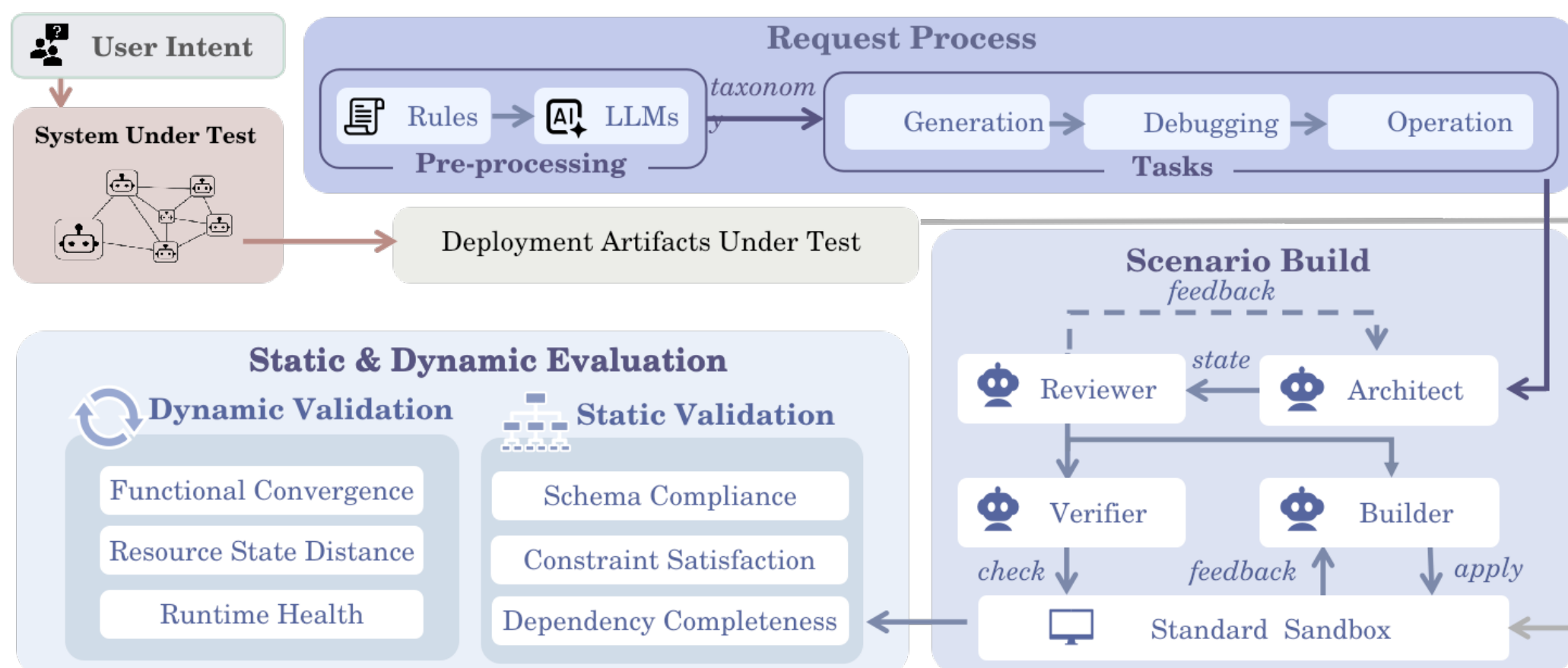
- Phase 1: Task Decomposition and Routing.** The Master Agent extracts context from the user’s problem and the *Cloud Native World Model*, breaking the task into granular subtasks and routing them to specialized domain agents (Config, YAML, Module).



VERA

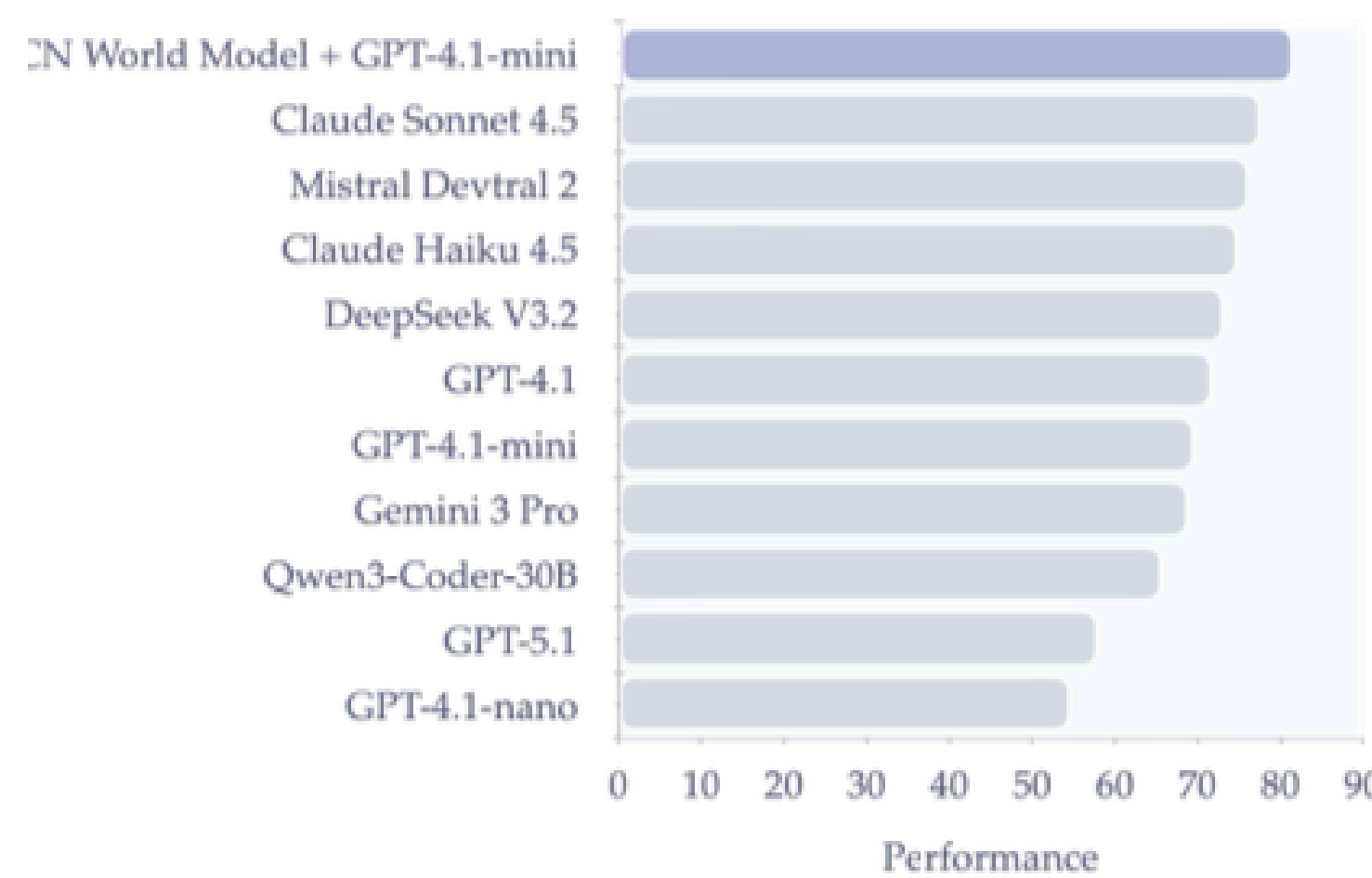


- Phase 2: Parallel Proposal Generation.** Domain agents independently generate configuration fragments. The YAML Agent ensures schema validity, the Config Agent handles environment setup, and the Module Agent structures code, while the Docs Agent injects in-line explanations for all outputs.
- Phase 3: Verifier-in-the-Loop Synthesis.** The system validates outputs using formal verifiers (e.g., kubeconform). Errors are classified and synthesized into minimum-edit patches stored as **Typed Memory Units (TMUs)**, ensuring monotonic reduction of schema violations without relying on ungrounded LLM self-critique.
- Phase 4: Response Generation and Dynamic Evaluation.** After successful system verification, the pipeline generates the corresponding YAML artifacts, which are then evaluated via dynamic evaluation to assess whether the candidate correctly transitions the environment from S_{pre} to S_{post} under execution and validation checks.



Experiment Result

We first evaluate baseline cloud-configuration solve rates on real-world Stack Overflow data using commercial models and supervised fine-tuning on Kubernetes knowledge documents. Anthropic’s Claude Sonnet 4.5 achieves the strongest baseline performance, but most models still struggle to reliably solve real-world cloud configuration tasks.



We then evaluate VERA on the same Stack Overflow-derived test set, where the current ground-truth coverage reaches 78.05%. With dynamic evaluation, VERA further improves deployment effectiveness, achieving an 83.2% YAML pass rate.